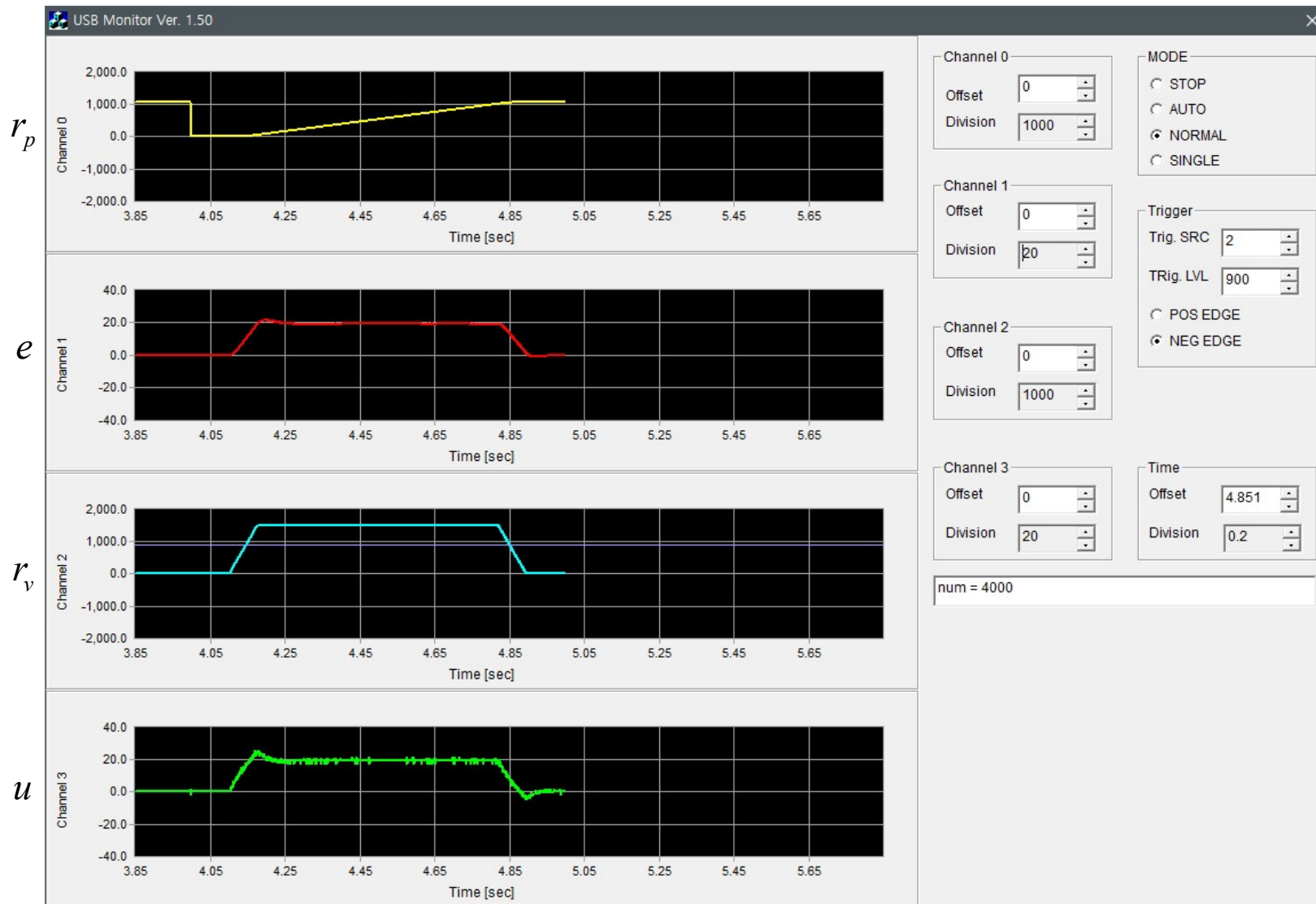
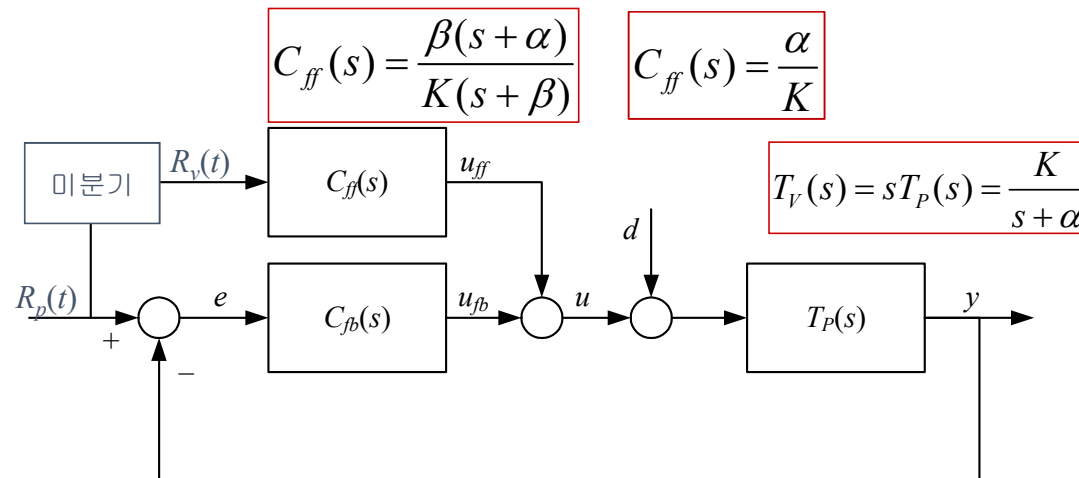
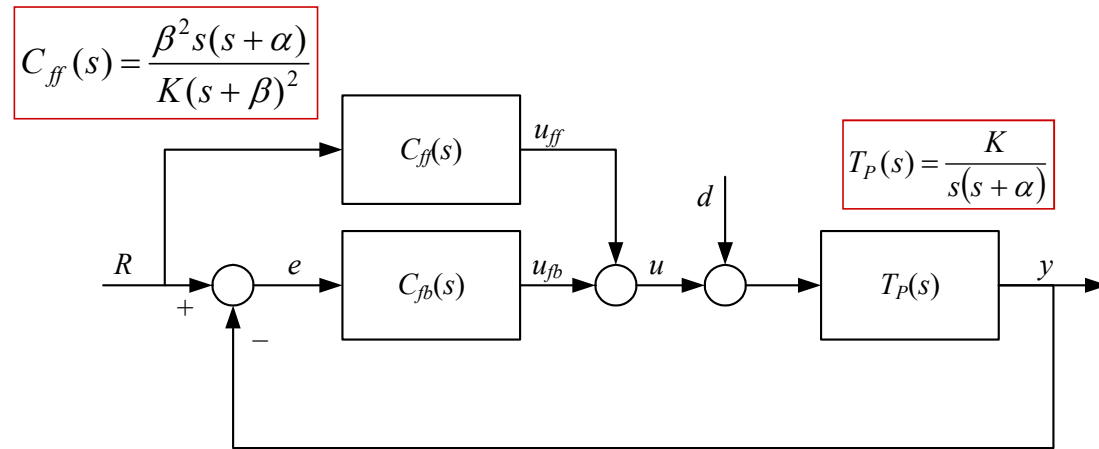


## No feedforward

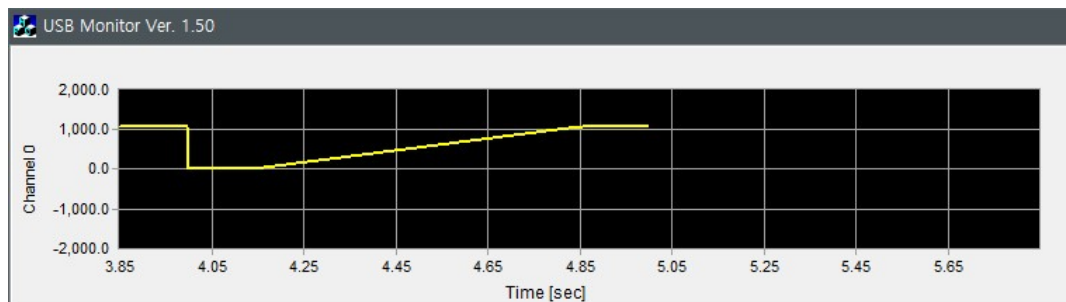




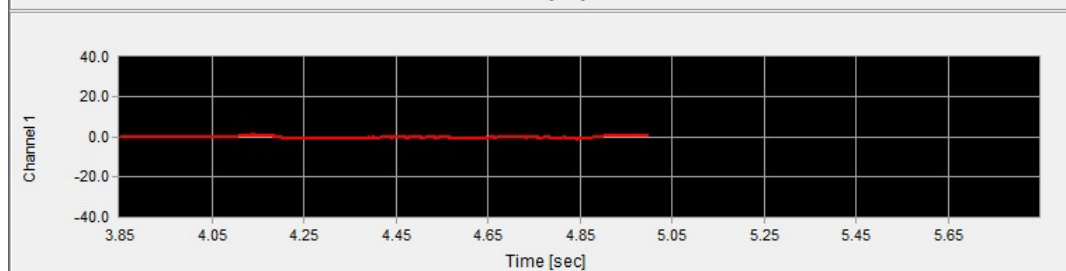
$$P_v(s) = \frac{K}{s + \alpha}$$

$$C_{ff}(s) = \frac{s + \alpha}{K} \frac{\beta}{s + \beta}$$

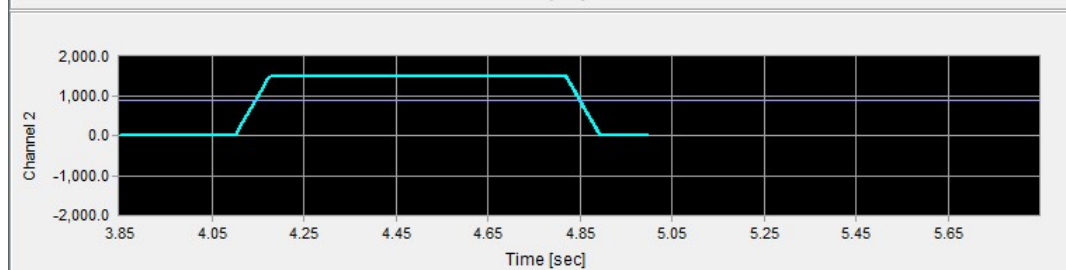
$r_p$



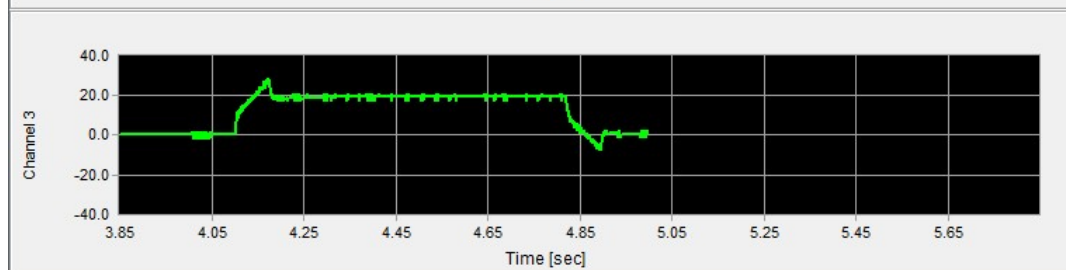
$e$



$r_v$



$u$



Channel 0  
Offset 0  
Division 1000

Channel 1  
Offset 0  
Division 20

Channel 2  
Offset 0  
Division 1000

Channel 3  
Offset 0  
Division 20

MODE  
☐ STOP  
☐ AUTO  
☒ NORMAL  
☐ SINGLE

Trigger  
Trig. SRC 2  
TRig. LVL 900  
☐ POS EDGE  
☒ NEG EDGE

Time  
Offset 4.851  
Division 0.2

num = 4000

$$P_v(s) = \frac{K}{s + \alpha} = \frac{78 \times 30}{s + 30} = \frac{2340}{s + 30}$$

$$\begin{aligned} C_{ff}(s) &= P_v^{-1}(s) \frac{\beta}{s + \beta} = \frac{s + \alpha}{K} \frac{\beta}{s + \beta} \\ &= \frac{s + 30}{2340} \frac{300}{s + 300} = \frac{0.1282s + 3.846}{s + 300} \end{aligned}$$



Tustin's method  
(bilinear transform)

$$C_{ffD}(z^{-1}) = \frac{0.1132 - 0.1098z^{-1}}{1 - 0.7391z^{-1}}$$

$$P_v(s) = \frac{K}{s + \alpha} = \frac{78 \times 30}{s + 30} = \frac{2340}{s + 30}$$

$$\begin{aligned} C_{ff}(s) &= P_v^{-1}(s) \frac{\beta}{s + \beta} = \frac{s + \alpha}{K} \frac{\beta}{s + \beta} \\ &= \frac{s + 30}{2340} \frac{300}{s + 300} = \frac{0.1282s + 3.846}{s + 300} \end{aligned}$$



Tustin's method  
(bilinear transform)

$$C_{ffD}(z^{-1}) = \frac{0.1132 - 0.1098z^{-1}}{1 - 0.7391z^{-1}}$$

$$P_v(s) = \frac{K}{s + \alpha} = \frac{2340}{s + 30} \approx \frac{K}{\alpha} = \frac{2340}{30} = 78$$

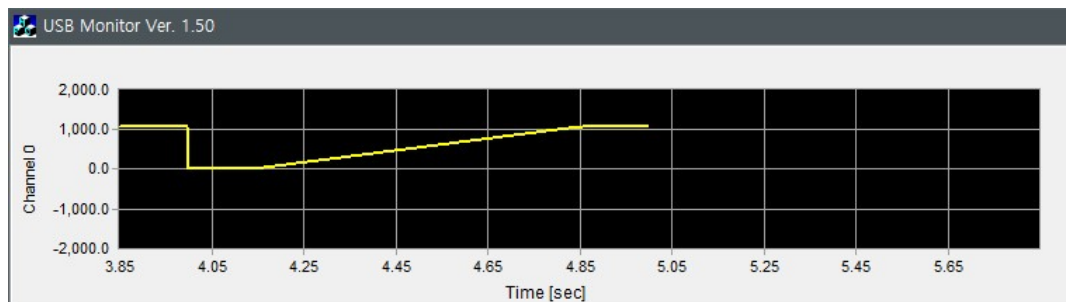
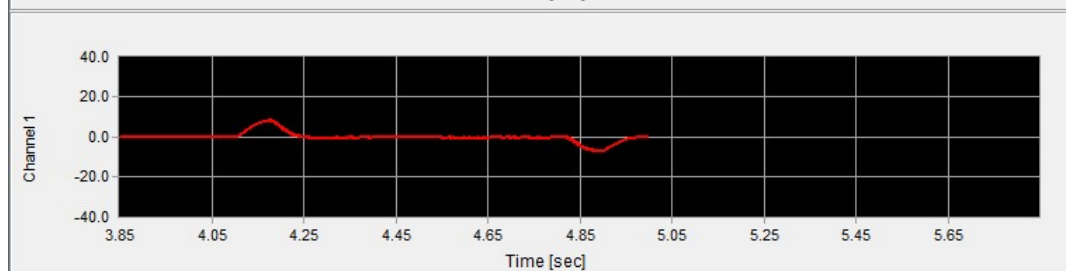
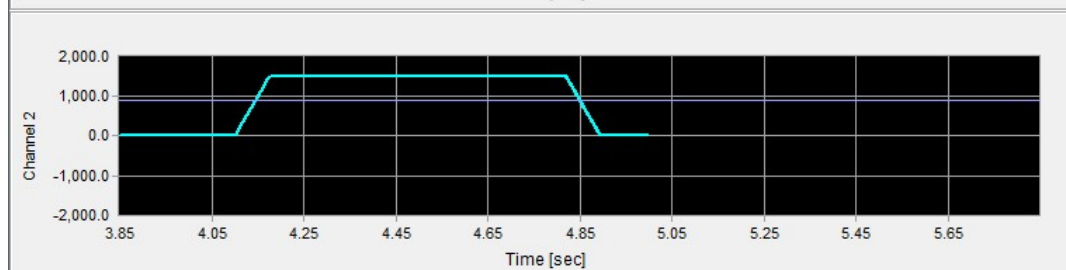
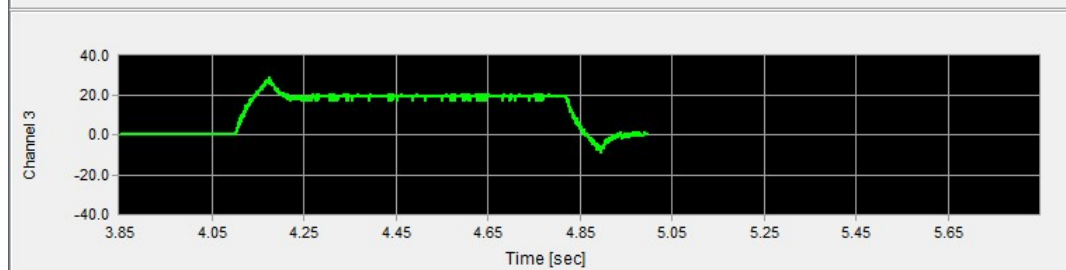
$$C_{ff}(s) = K_{ff} = \frac{\alpha}{K} = \frac{30}{2340} \approx 0.013$$



$$C_{ffD}(z^{-1}) = 0.013$$

$$P_v(s) = \frac{K}{s + \alpha}$$

$$C_{ff}(s) = K_{ff} = \frac{\alpha}{K}$$

 $r_p$ 

 $e$ 

 $r_v$ 

 $u$ 


USB Monitor Ver. 1.50

Channel 0  
Offset 0  
Division 1000

Channel 1  
Offset 0  
Division 20

Channel 2  
Offset 0  
Division 1000

Channel 3  
Offset 0  
Division 20

MODE  
☐ STOP  
☐ AUTO  
☒ NORMAL  
☐ SINGLE

Trigger  
Trig. SRC 2  
TRig. LVL 900  
☐ POS EDGE  
☒ NEG EDGE

Time  
Offset 4.851  
Division 0.2

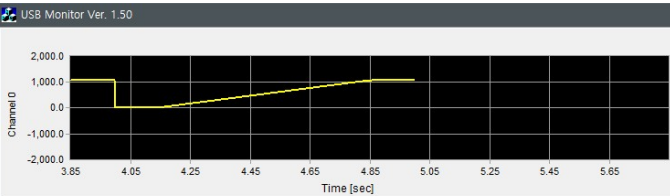
num = 4000

No feedforward

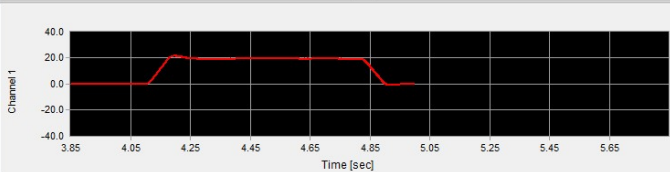
$$C_{ff}(s) = K_{ff} = \frac{\alpha}{K}$$

$$C_{ff}(s) = \frac{s + \alpha}{K} \frac{\beta}{s + \beta}$$

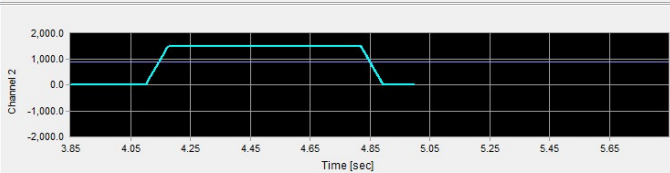
$r_p$



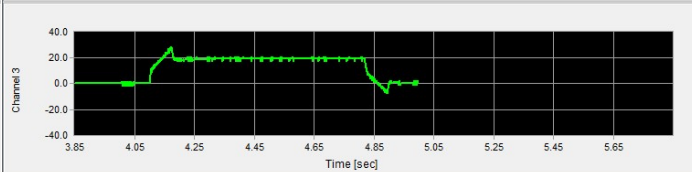
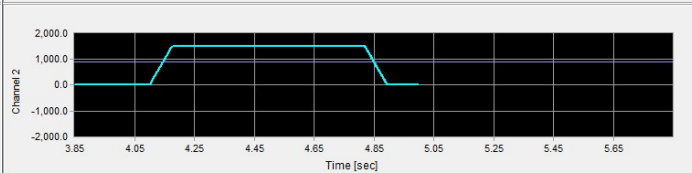
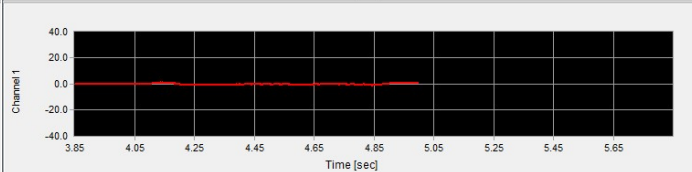
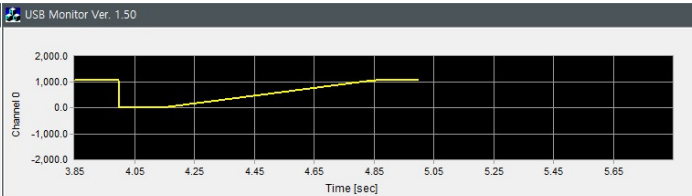
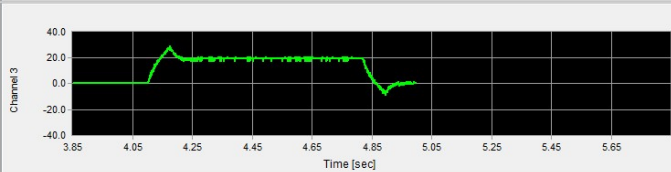
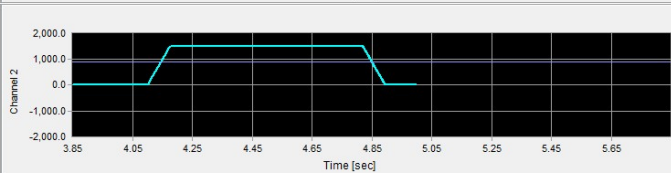
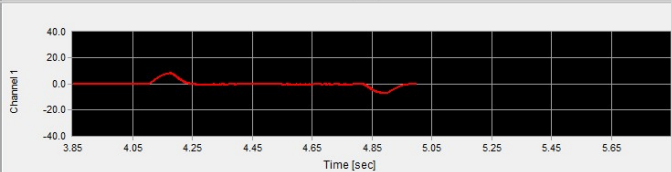
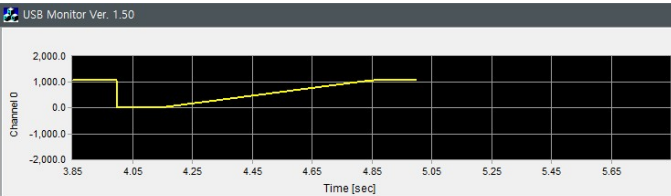
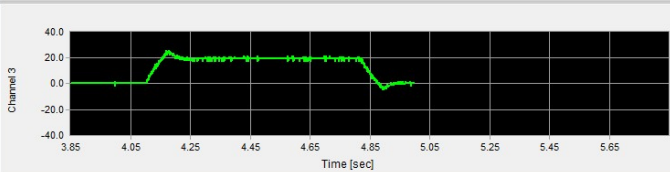
$e$



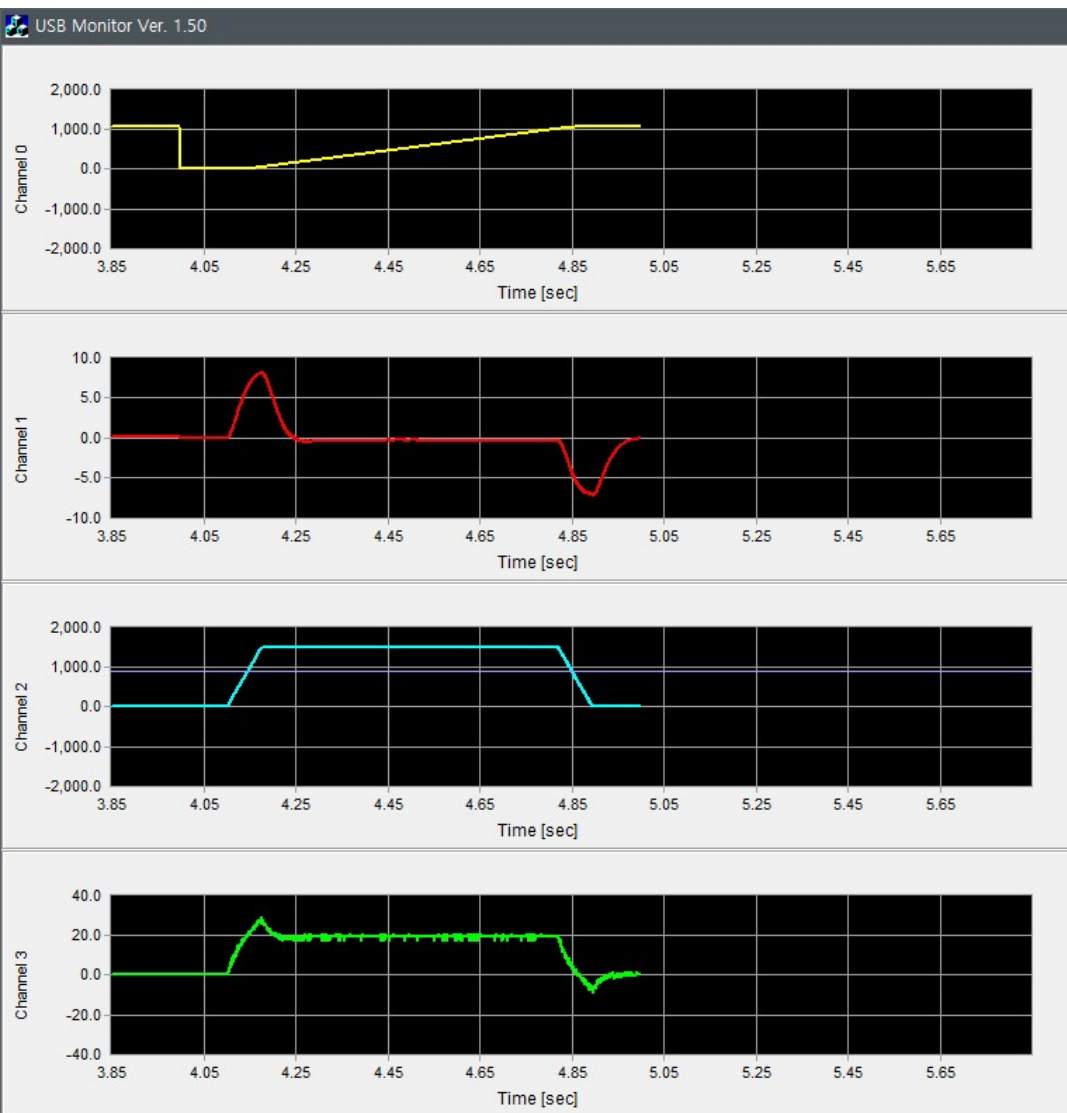
$r_v$



$u$

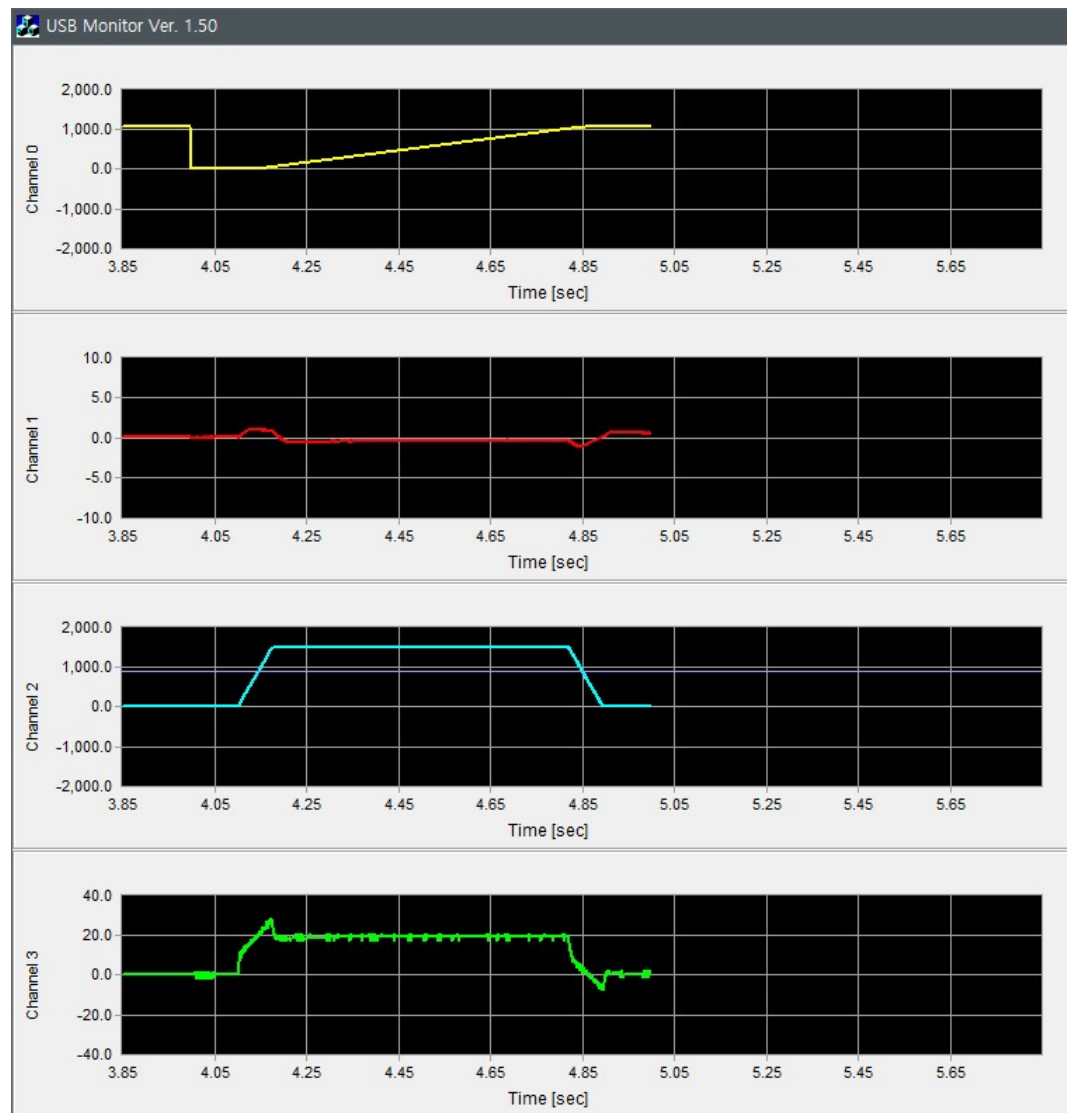


$$C_{ff}(s) = K_{ff} = \frac{\alpha}{K}$$



$$C_{ff}(s) = \frac{s + \alpha}{K} \frac{\beta}{s + \beta}$$

$r_p$



$e$

$r_v$

$u$